

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 6

The digital divide

17 questions from 17 question papers, May/June 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

- [6]

6 The digital divide affects many people. There can be a divide between older and younger people. Describe how this digital divide could be reduced.

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10 Describe, listing the groups affected, what is meant by the digital divide.

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Please turn over for Question 11.

[5]

4 The digital divide can exist between people in less industrially developed nations and those in more industrially developed nations.

(a) Explain why this digital divide exists.

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(b) Describe the effects of this form of digital divide.

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6 Describe how a local government can help reduce the digital divide in rural areas.

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6 Describe, by referring to **two** affected groups, what is meant by the digital divide.

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14 Identify **two** groups of people who can be negatively affected by the digital divide. For each group, suggest a way the digital divide can be reduced.

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13 Describe the effects of the digital divide on health services.

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Mark schemes

Mark scheme · 9626/11 May/June 2022 · Question 9

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Question	Answer	Marks
9	Six from: Infrastructure for broadband is less developed in rural areas than in cities... ...any existing infrastructure tends to use a lower level of technology making it difficult to install high speed broadband In rural areas people tend to be further away from a cabinet causing longer cables to be used leading to slower speeds Rural areas can have difficult terrain which make it difficult for companies to develop the infrastructure/lay fibre cables Rural parts tend to have a lower concentration of people/population so providers cannot recuperate costs quickly/less willing to invest in infrastructure More expensive for providers to offer services in rural areas and the extra costs are passed onto the customers Cities can have more providers competing to provide broadband than rural areas making it cheaper for the customer In rural areas people can use satellite broadband but it tends to be slower than cabled broadband connections Satellite broadband can be more expensive than cabled broadband connections Can use smartphones but signal quality depends on the number of towers and in rural areas there are fewer	6

Mark scheme · 9626/11 Oct/Nov 2022 · Question 6

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Question	Answer	Marks
6	Five from: e.g. • Younger people could train older people (1) • The government could subsidise internet access for older people (1) • Government could encourage libraries/community centres to offer free internet access to older people/those who do not have access (1) • Retirement villages could be built incorporating technologies (1) • Information providers should be required to produce content in suitable formats so that people with weaker eyesight can easily access the content (1) • Grants can be made to organisations to deliver digital literacy training programmes to older people (1) • Schools could allow older people access to technology facilities in the evenings//during school holidays (1) • Schools could provide teaching and training to older people to improve their skills (1)	5

Mark scheme · 9626/13 Oct/Nov 2022 · Question 10

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Question	Answer	Marks
10	Six marks available: - The gap between people who have access to and the resources to use new information/communication technology and those who don't ... (1) ... This technology can include the telephone, television, personal computers and the Internet (1) - The quality of connection to the Internet may vary between groups/countries (1) - Differing levels of infrastructure between (e.g.) countries (1) - The price of connection to the Internet may vary between groups/countries//ability to afford access to the internet (1) - May refer to inequalities between individuals, households, businesses, or geographic areas (minimum of two mentioned) (1) - The divide between those who have the skills/knowledge/ability to use the technologies and those who do not (1) - The digital divide can exist between those living in rural areas and those living in cities (1) - The digital divide can exist between the educated and uneducated (1) - The digital divide can exist between socioeconomic groups/rich and poor (1) - The digital divide can exist between old and young people (1) - The divide between more and less industrially developed/technologically aware nations (1) - The divide between countries or regions of the world is referred to as the global digital divide (1)	6

Mark scheme · 9626/12 Feb/March 2023 · Question 8

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Question	Answer	Marks
8	FIVE from: - Some people in developing countries have no actual postal address making the delivery of goods which are purchased online difficult (1) - Many people in developing countries have no access to broadband so speed of purchase is slower (1) - Many people in developing countries have no PCs/laptops/smartphones/tablet computers so need to borrow a PC/go to a library (1) - In developing countries power outages can mean that connection is lost during the process and so have to start again/lose the sale (1) - Many people in developing countries do not have bank accounts and credit cards so have to organise for someone else to buy the product on their behalf (1) - In many developing countries, goods bought using the internet are not covered by consumer-protection laws so people have no protection from unscrupulous sellers (1)	5

Question	Answer	Marks
4(a)	Three from: • The quality of connection to the Internet may vary (1) • Speed of access to the Internet may vary (1) • Some countries may not have a well-developed broadband (infrastructure)/rely on dialup connections (1) • Lack of incentive to invest in the country/nation (as a whole) (1) • In less developed countries they might have less infrastructure (may exemplify) (1) • Not all countries are able to keep up with the constant changes in technology (1)	3
4(b)	Three from: • The digital divide gets wider (1) Award ANY three implications of a digital divide: For example • higher/lower quality on education (1st) • higher/lower quality on health (1st) • higher/lower quality on access to resources (1st) • higher/lower quality on lifestyle (1st) • higher/lower quality on opportunities (1st) • higher/lower quality on work/employment opportunities (1st) • higher/lower quality on trade (1st) • reduced access to social media (1st) • higher/lower access to information (1st) A further mark can be awarded for an explanation of the impact (1) A final mark may be given for the alternative comment – such as MP3 in the example below. Note: can ONLY be given as a 3rd mark. E.g., reduced access to education (1) means lower skill level (1) whilst those in industrial societies have better access (1)	3

Mark scheme · 9626/12 May/June 2023 · Question 7

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Question	Answer	Marks
7	Five from: • Different socioeconomic levels imply differing levels of (disposable) income (1) • Different incomes/socioeconomic groups impact on what people can buy/access/afford (1) such as 'things' ... (must be an extension linked to technology) (1) such as acquisition of skills/paying for training (1) • This impacts on what people can commit to renting (1) • New equipment is costly (1) • Old equipment is not as advanced (1) • People from lower socioeconomic groups tend to have lower digital skills (1) • No/reduced incentive for companies to develop cheaper systems ... (1) ... because the profits would be lower (1) • No/reduced incentive to provide infrastructure to poor areas (1) ... because reduced demand in the area (1)	5

Mark scheme · 9626/13 May/June 2023 · Question 7

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Question	Answer	Marks
7	Six from: e.g. • Young people have grown up with the technology/have only ever known the use of smart phones/internet //its seen as part of their everyday life (1) • Older people may feel they are too old to learn about new technology (1) • Younger people are more likely to take risks/older people are more risk averse (1) therefore young buy new technology/old do not buy new technology (1) • Many older people already have established means of communication and so do not see the need to explore new technology (1) • IT businesses tend to aim new technologies at younger people ... (1) ... companies make few modifications to help older people use them more easily (1) • Technology seen as something for the young (1) • Older people can be awkward with their hands and fingers making the use of smartphones a problem (1) • Elderly people also find it difficult to read text on a screen or use a keyboard (due to visual impairments) (1)	6

Question	Answer	Marks
6(a)	Four from: A general statement of impact: • Inability to find information due to inability to access search engines (or other example of cause. Example must be more than “cannot access the web”) (1) • Tasks take longer to complete, which wastes time (1) Up to four marks for four effects and explanations. Areas include: Reduced health care, e.g. • Inability to get onto a waiting list as cannot access doctor’s website (1) • Not aware of imminent health risks as do not have access to a national news service (1) Reduced ability to work remotely, e.g. • Unable to have face-to-face communication over the web, and so cannot work remotely (1) Impact on education, e.g. • Cannot access online learning resources, and so quality of education is lower than could be (1) • Unable to do wider research and so will lack wider understanding of the topic (1) Impact on purchasing, e.g. • Inability to compare prices of products as cannot access search engines (1) • Inability to check on availability of products as can not access search engines (1) Other areas include, but are not limited to: • Impact of limited access to online facilities, banking • Impact of limited access to communication • Impact of limited access to mental health • Impact of limited access to online entertainment	4
6(b)	Three from: • Build/improve access to the internet/faster broadband (in rural areas) (1) • (Governments/industry can) provide/improve the needed/broadband infrastructure (for fast broadband in rural areas) (1) • Government provides incentives/subsidise to industry to build infrastructure (1) • Government investigates alternative means of distribution (1) • Create cyber/internet cafes (in rural areas) (1) • Deliver access using satellite technology (1)	3

12	Six available. One for a definition of the digital divide: - The digital divide is the gap between those who have access to//use technology and those who do not Max six from: - In this case, the gap is caused by differing levels of knowledge about what technology is available (1) AND differing levels of knowledge about what technology can achieve//benefits to be gained (1) - Not everyone is able to keep up with/keep aware of the constant changes in technology - A person who is unaware of what is available//what can be achieved will not use //not know how to use technology - A person who is aware of what is available//what can be achieved will use technology - Increased digital awareness leads to increased/better infrastructure// more investment in IT (accept personal equivalences) (1) which then leads to increased digital awareness (1) (accept reverse argument for digital unawareness) This divide may be reduced by education/research/experience - This divide may exist within any socioeconomic bracket - This divide may exist within any age group.	6
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Question	Answer	Marks
3	Six from, e.g.: One mark for a definition of the digital divide: e.g. - The digital divide is the gap between those who have access to//use modern/ICT technology and those who do not (1) Improve communications coverage/access to the internet/access to broadband (1) Improve infrastructure (1) such as (e.g.) invest in satellite technology/build more mobile phone masts/lay fibre cabling (1) - Provision of public access to internet facilities in, e.g. (ONE OF) government buildings/libraries/schools and internet kiosks/cafes (1) - Provide/subsidise laptops/PCs/technological equipment to, e.g. (ONE OF) the elderly/the poor/school children (must be a specific group likely to be negatively affected by digital divide) (1) - Provide/subsidise of mobile or cell phones to, e.g. (ONE OF) the elderly/the poor/school children must be a specific group likely to be negatively affected by digital divide) (1) - Provision of training services/sessions (1) - Subsidise access to a named service (1)	6

Mark scheme · 9626/12 Oct/Nov 2024 · Question 5

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Question	Answer	Marks
5	SIX from: - Highly educated people are often employed in jobs that offer exposure to IT (1) - Highly educated people will often undertake IT training as part of their education//have IT skills as a result of their education (1) - Highly educated people are often aware of IT/importance of IT/benefits of IT/possibilities of IT (1) - Highly educated people have more access to the higher paid jobs (1st) therefore can afford/access the (e.g.) higher speed internet/most up to date IT equipment (1) (accept converse argument) - People with lower levels of education often work longer hours, therefore have less time to access IT (1) - Lower educated people are less like to invest in IT (1)	6

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Question	Answer	Marks
8	EIGHT from, e.g.: Award ANY description of differing access to Information Technology leading to digital divide. ONE mark per description e.g. - Access to fast broadband can be better for those people living in urban areas than for those living in rural areas (1) - The quality of connection to the internet may be poorer in underdeveloped countries (1) - Buying televisions/personal computers/smartphones can be more expensive in rural areas (1) - Richer people are more/poorer people less able to afford the latest versions of televisions/personal computers/smartphones computers (1) - Lower paid workers may not be able to afford the hardware to be able to access the IT (1) - Uneducated people are often not aware of IT/importance of IT/benefits of IT/possibilities of IT (1) - Young people have grown up with the technology/older people may feel they are too old/unwilling to learn about new technology (1) - Disabilities of some users may inhibit the use of smartphones/computers. (1)	8

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Question	Answer	Marks
6	One mark per valid description to a maximum of four marks. E.g.: • Expand broadband access/improve infrastructure. (1) • Establish/support community centres with computers/internet access. (1) • Offer workshops/courses to teach residents computer skills/ICT (1st) – such as – accept any suitable use of ICT (online or offline) (1) • Educate residents on//raise awareness of any identified aspect e.g. cybersecurity/online safety/technology. (1) • Establish equipment loan programs (1) • Equip schools with improved/better/latest technology (1) • Subsidise access to internet (1) • Subsidise purchase of computer equipment (1)	4

Mark scheme · 9626/11 May/June 2025 · Question 6

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Question	Answer	Marks
6	Two marks from: • The gap between people who have access to//resources (computer hardware) to use new information//communication technology and those who don't (1) • The gap between people who have access to information and those who don't (1) Max 2 from: People with/from: • Varying ICT skills (1) • Living in rural areas and those living in cities (1) • Different education levels (1) • Different socioeconomic groups (1) • Different age groups (1) • Different industrially developed/technologically aware nations (1)	4

Question	Answer	Marks
14	Four marks for two matched group-remedy pairs: e.g. Groups (1st) • Old people (1) • People in rural areas (1) • Low/different levels of education (1) • Lower socioeconomic group//people who cannot afford to purchase (1) • People with learning difficulties (1) • People with physical impairments (1) • People with sensory impairments (1) • People in less industrially developed nations (1) Remedies (1) • Education/training (1) • Drop-in centres support (1) • One to one support (1) • Invest in infrastructure (1) • Provide specialist interfaces (1) • Provide (cheap) loans/subsidies/price ceiling/free (1) • Recycling/refurbish/repurpose equipment (1)	4

Question	Answer	Marks
13	Six from: e.g. • Can give differing levels of access to health services (1) • Increased risk of death/length of illness extended (1) • May not have online health services (1) – Reasonable implication (1) • Delayed access to results (1) • May need to attend doctor in person/travel to hospital (1) • Not everyone may be able to book online (1st) – need to visit/telephone (1) – which means time is wasted/used less efficiently (1) • Not everyone can order/reorder medicine/prescriptions online (1) • Doctors can work remotely from the patient (1st) – increased access to doctor/patient (1) – ambulances are less frequently used (1) • Not everyone can submit/receive test results online (1) • Not everyone can be monitored remotely (1)	6

Question	Answer	Marks	Guidance
7	EIGHT from: - Because there is a digital divide (between them) (1) Possible themes include: - Different locations (1) with different access (1) - Lower disposable income/affordability/socio-economic difference (1) affecting access (1) - Different levels of fear (of any type)/comfort (1) however, this fear will quickly disappear with use (1) if issue is not addressed, this problem will get worse (1) - Different day to day experience/training (1) - Different jobs make different use of IT (1) - Disability (1) - Lack of awareness of possibilities (1) - Any example of choice or lifestyle (absence of or different need lack of interest, lack of online friends (1) Answers may be awarded for: Identification of issue (1) Explanation/description of issue (NOT just another version of the same issue) (1)	8	